

Perpetual Futures in the Cryptocurrency Market:

The Role and Impact of Funding Rates

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Foundations of Digital Money & Blockchain

Money is a verifiable document, created to measure, store, and exchange value without the need to barter. The form in which money has been used has changed drastically in the past 200 years, & with these changes, new inventions and technologies are being created. Traditional money, the form used every day, relies on tightly managed centralised systems.

It comes in two forms, viz. Commodity Money and Fiat Money:

1. Commodity Money:

Commodity money derives value from the intrinsic worth of the material from which the asset is made from. These commodities were used as money because they were difficult to counterfeit, universally recognisable, and scarce (preventing inflation). Examples include gold, silver, copper, &c.

2. Fiat Money:

Fiat money is a government-issued currency (a tangible form of money in circulation) whose value does not derive from its intrinsic worth, but rather from supply and demand forces, public trust, and the stability of the government that issues it. They circulate because the participants in the economy accept it, and it is supported by government policies like taxation, monetary policy, and regulation in supply. Examples include the US Dollar, the Euro, the Rupee, &c.

Banks maintain a ledger, which essentially contains records of financial transactions, that has information on how value moves between two parties. Historically, ledgers were maintained on paper and were used by merchants, bankers, governments, &c. In the modern day, ledgers are maintained electronically, in a centralised database. For example, your bank balance on an

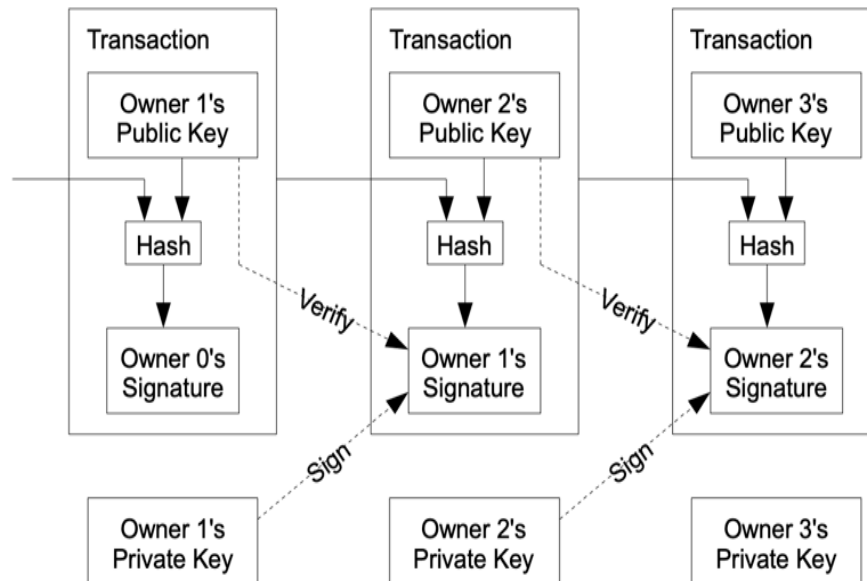
app is an electronic ledger, and credit card statement records are also electronic ledgers. As society began to digitalise, the physical nature of money has largely started to become impractical, and most modern transactions don't involve physical notes or coins. As banks, financial institutions, and payment networks started maintaining digital ledger entries, money essentially has become a purely informational record stored in centralised databases and has become digital in nature. This centralised ledger system works, but it also means every transaction, correction, permission, and approval depends on a single institution's integrity, infrastructure, and policies. Another important point in centralised finance is trust in the intermediary. Intermediaries extract a profit from shortening the time in which buyers and sellers match for a transaction. You hand over control of your assets during the process of a trade. If you want to regain control over your assets, it requires trust and faith in the intermediary that they are able and willing to hand over control back to you.

In the late 2000s, an idea for a financial system that reduces the need for intermediaries arose. The goal of this system is the decentralisation of the custody of assets (from a bank, exchange, &c.) during the transaction process, and to allow users to transact among one another without the presence of a middleman who could block transfers, freeze accounts, or restrict withdrawals. This system became known as decentralised finance or DeFi for short, which had the aim of removing the power that central financial institutions hold.

For this system to work, it requires another type of payment ecosystem that would use new protocols for transactions, rather than the traditional centralised system, which is heavily reliant on an intermediary. The technology for this protocol came in the form of the blockchain.

In October 2008, an anonymous figure using the alias Satoshi Nakamoto published a 9-page research paper titled "Bitcoin: A Peer-to-Peer Electronic Cash System." This document proposed a public digital ledger that was visible to anyone and was controlled by no single authority, which was secured entirely

through cryptography. This digital ledger became known as the blockchain, with Bitcoin being its first application.



The way in which money is transferred through a chain of digital signatures. Every owner transfers assets by digitally signing a 'hash', which contains the payee's public key and the previous transaction data of this asset. The payer uses their private key to make a digital signature of this hash, which proves that the payer chose to transfer this asset to the payee. The hash is also timestamped; that is, the time at which the hash was created is recorded in it. This proves that each transaction must have happened for the hash to be accessed.

To prevent someone from signing multiple hashes and essentially double-spending their asset, all transactions are publicly announced, and all nodes (a computer that is connected to a cryptocurrency network) that receive info about transactions are required to decide on the transaction history for this asset. The payee needs proof that most nodes agree on a single transaction history, too.

But who adds these transactions onto the blockchain? For Bitcoin, blocks of transactions are added to the blockchain after approval from a consensus

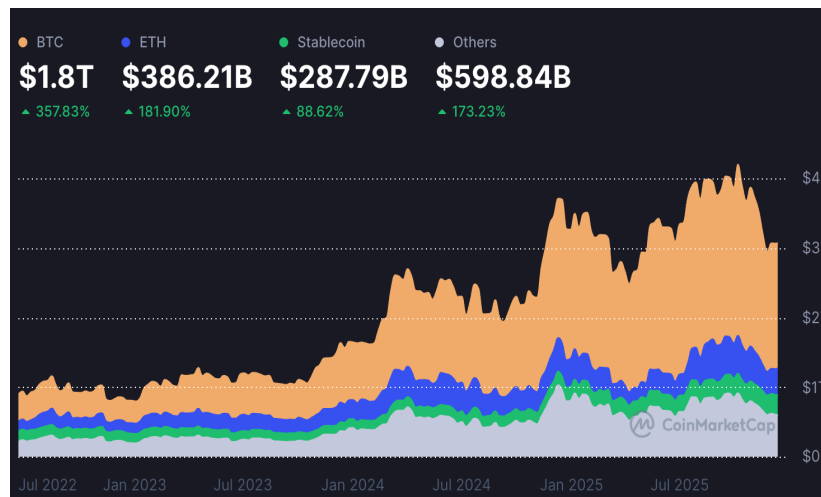
mechanism called proof-of-work (PoW). Users called miners, who solve highly computationally intensive cryptographic puzzles, are allowed to authenticate the transactions that take place. This process is highly resource-intensive, so regular users can't solve PoW puzzles on their computers to validate their transactions. As an incentive for solving these puzzles and validating other users' transactions, miners are rewarded with bitcoin, and this is how bitcoin's supply is distributed. But to control the supply of Bitcoin, the reward is halved every 4 years. Currently, the bitcoin block reward is 3.125 BTC, and around the year 2140, the reward is projected to be zero. Blockchain technology isn't only used for cryptocurrency ledgers; it has a vast number of applications, including supply chain tracking, tokenisation, and many more.

Where does Bitcoin's value derive from? Satoshi Nakamoto put a hard cap of 21 million Bitcoin that can be mined. That means no more Bitcoin can be created beyond this hard cap. Right now, around 2-3 million Bitcoins are actively circulated in markets. It also costs resources to mine Bitcoin for solving the Proof-of-Work (PoW) puzzle. So, there is a base cost to mine Bitcoin. Since nobody owns Bitcoin, via decentralisation, it means that Bitcoin is resistant to monetary inflation. The blockchain technology behind Bitcoin is extremely secure. The use of public and private keys and the linking of transactions via timestamped hashes is what makes the blockchain ledger system so safe. Broadly, Bitcoin has gained value from investors due to its scarcity, the capital-intensive nature of its extraction, its security, and its uniqueness from other forms of currency around the globe. Unlike Bitcoin, Ethereum, another cryptocurrency, does not have a maximum supply cap, yet its value is shaped by other forces like network usage, transaction fees, staking rewards, and the growth of decentralised applications.

Cryptocurrency Market Essentials

Cryptocurrency Markets are 24/7, decentralised, and global, where participants can buy, sell, and trade various crypto-assets like Bitcoin, Ethereum, &c.

Cryptocurrency markets are different from traditional financial markets, as they lack a central authority like a government or a bank to regulate these transactions. In crypto markets, price formation is influenced not only by supply and demand but also by a variety of technological factors, including network congestion, miner or validator incentives, automated high-frequency trading bots, and the behaviour of decentralised protocols.



The currency market capitalisation of the cryptocurrency market is roughly 3.1 trillion USD. For reference, that is ~10.1% of the global gold market cap. Bitcoin dominates the crypto market, accounting for ~60% of the market cap, followed by Ethereum at ~13% of the market cap.

This is largely due to the fact that BTC and ETH had a first-mover advantage, being some of the oldest crypto-assets available, along with the fact that BTC and ETH are now being institutionally adopted in the form of spot ETFs and a growing perception of being safer and more risk-free than other alternative coins.

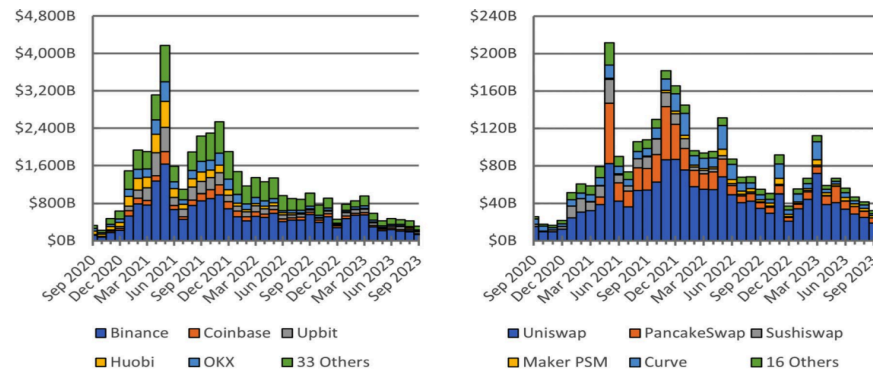
Another important type of cryptocurrency is the stablecoin. A common perception of crypto-assets is that they are very volatile. Everyday financial transactions would be difficult if the value of say 1 BTC fluctuates abruptly. Stablecoins were devised to maintain a stable, less volatile value compared to other crypto-assets, based on the principle of anchoring their value to fiat currencies or commodities. A stablecoin issuer can anchor the value of the

stablecoin to the value of a fiat currency via multiple methods. One could be by holding reserves of fiat currencies (for a USD-based stablecoin, that would be instruments like US treasuries, cash, &c.) equivalent to the value of the stablecoins issued. Another way is to do the same, but using crypto-assets as collateral. Another type, algorithmic stablecoins, peg their value to fiat currencies by supply and demand dynamics. If the price of a stablecoin rises above 1 USD, new stablecoins will be minted, and if the price drops, the supply of the stablecoin is reduced by incentivising 'bond' tokens with the stablecoins.

Stablecoins still aren't perfectly stable, and there have been periods where the value of popular stablecoins has crashed. For example, TerraUSD, an algorithmic stablecoin, crashed to 10 cents and wiped out 45 billion USD worth of market capitalisation in a week.

Since cryptocurrencies are not governed by any central authority, any participant can perform transactions on the network. Users and bots could flood the network with spam transactions, overwhelming the mempool (a waiting area for unconfirmed transactions over the blockchain) and making transaction validation inefficient. To tackle this, cryptocurrencies use transaction fees called 'gas fees' to prevent spam and to pay for all the resources, such as storage, computational resources and miners to process the transaction.

Cryptocurrency exchanges are the platforms that facilitate the aforementioned buying, selling & trading of crypto-assets in the crypto market. They are privately owned and facilitate exchanges between crypto-assets, fiat money, NFTs & more. Crypto exchanges can be broadly divided into two main types, viz. Centralised Exchanges (CEX) & Decentralised Exchanges.



Centralised Exchanges operate as financial intermediaries between buyers and sellers, similar to how stock exchanges operate. They allow users to buy and sell crypto-assets via spot orders (buying or selling at the current price) & limit orders (buying or selling when the price hits a certain target price). They operate using an order book system, similar to the stock exchange, where an algorithm matches users based on their order price in the most efficient way possible. When a user places an order, the exchange first verifies that the user has sufficient balance and meets all compliance requirements. Once validated, the order enters the exchange's matching engine. This engine maintains the order book and matches buy orders with sell orders based on price-time priority. Market orders are executed immediately against available liquidity, whereas limit orders rest in the order book until a matching counter-order is found.

Importantly, centralised exchanges do not execute every trade on the blockchain. Because executing trades on-chain would be slow and costly, all trading activity occurs within the exchange's internal infrastructure. The exchange updates its internal ledger to reflect balances after each execution. This design allows centralised exchanges to provide fast execution speeds and deep liquidity, but also results in variations in price across exchanges, as each exchange runs an independent internal market. Arbitrage bots monitor these price differences and trade across platforms to reduce price discrepancies.

The financial intermediaries, by offering user-friendly, reliable, 'safe' crypto-assets and not allowing fraudulent crypto-assets from their platform, and

allowing users to leverage their investments (borrowing money to amplify returns), charge commissions and transaction fees. CEXs are large potential targets for hackers; for example, in 2014, the CEX Mt. Gox was the victim of a theft of 850,000 BTC. CEXs also have the disadvantage of slippage (commissions & transaction costs) that could be very large for high-volume transactions. Examples of popular CEXs include Binance, Coinbase, Kraken, and many more. Decentralised Exchanges allow users to transact crypto-assets directly (peer-to-peer) without the need for an intermediary. Decentralised exchanges operate on fundamentally different principles. A DEX is a blockchain-based trading platform that allows users to trade cryptocurrencies directly from their wallets. Unlike centralised exchanges, DEXs cannot convert traditional fiat money into digital assets; instead, users must already possess blockchain-native assets in self-custody wallets. A wallet is defined by ownership of a private key, which authorises transactions on the blockchain. The DEX acts only as a protocol that facilitates the connection between the user and the blockchain.

Smart contracts govern all DEX operations. These contracts automate trade execution, calculate slippage, apply transaction fees, distribute rewards to liquidity providers, and enforce protocol rules. Because smart contracts are permissionless, users retain full control over their assets until the moment of execution. This non-custodial model eliminates the risk of exchange insolvency (bankruptcy) or withdrawal freezes but introduces new risks, including smart contract vulnerabilities, impermanent loss for liquidity providers, reliance on blockchain throughput, and exposure to front-running bots that exploit public mempool visibility.

The way DEXs allow users to trade crypto-assets is through an algorithm called Automated Market Makers (AMMs). AMMs work via smart contracts called Liquidity Pools, which hold two different types of crypto-assets (like BTC and ETH). Users called liquidity providers deposit an equal amount of both types of tokens into the liquidity pool and earn trading fees because they deposited their assets in the liquidity pool. Most AMMs use a constant product formula ($x*y=k$), in which the product of the reserves of both tokens equates to a constant. This

prevents the liquidity pool from running out of tokens and also allows the price of the assets to adjust automatically after every trade. Bigger trades affect the price more, and smaller traders affect the price less. When you buy say BTC from a BTC-ETH pool, you add ETH to the pool and remove BTC from the pool. The ratio of BTC-ETH changes, and hence the price changes automatically.

DEXs are often very complex and only allow users to trade between different crypto-assets, and not with fiat money. This prevents users who don't have crypto-assets from using DEXs. Another issue with DEXs is that they largely lack

Aspect	CEX	DEX
Risk of misappropriation of funds in transit	High	None
Exclusion of participants	Feasible	Unfeasible
Direct trading costs	Prevalent	None
Trading partner discovery	Trivial	Complex
Exchange rate transparency	Transparent	Opaque
Opportunity costs due to tied capital	Low	High

volume, and hence, a liquidity issue arises. Around 99% of crypto trade volume occurs on CEXs, so difficulties in finding buyers and sellers can arise.

Furthermore, because AMM pricing is determined by mathematical formulas rather than order books, large trades often face significant slippage. DEXs remove the custody of an intermediary and remove the unnecessary slippage that arises when trading through these intermediaries. Popular DEXs include Uniswap, dYdX, Kyber, and many more.

Arbitrage

Arbitrage is the practice of extracting a profit by taking advantage of price differences of the same financial instruments in different markets. More specifically, arbitrage is a transaction that has no possibility for a negative cash flow whatsoever, but has at least one possibility for a positive cash flow, and hence is the possibility of extracting a risk-free profit. Price differences exist in markets due to inefficiencies in the markets. These inefficiencies appear as information asymmetry in different markets, transaction costs, and supply and

demand forces that are not symmetric in all markets. Additionally, in electronic trading, latency in price updates is a notable issue.

The principle of “No Arbitrage” states that in efficient markets, risk-free profits (arbitrage) aren’t possible to obtain. This is because if risk-free profits were obtainable, then they would be instantly exploited by market participants, causing the price difference (profit) to converge (due to supply and demand effects), effectively removing the profit. Arbitrage ensures the price of a financial instrument reflects all available information, thereby removing existing risk-free profit opportunities, reducing the likelihood of future arbitrage.

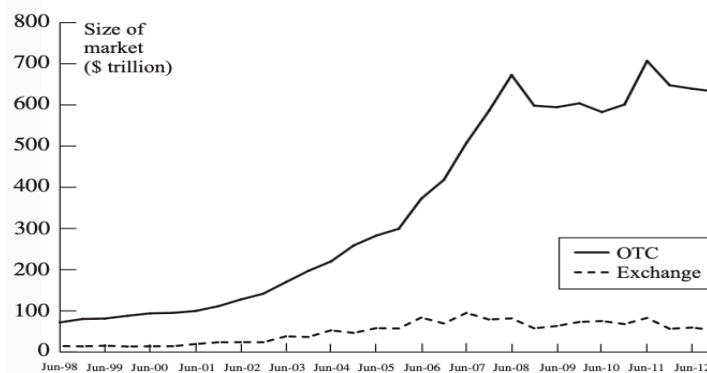
A Deep Dive Into Derivatives

Consider a basic agreement involving a bicycle, where a buyer and a seller enter into a contract today, which states that the cycle will be bought three months later for a fixed price of ₹500. When the agreement is made, the bicycle may also be ₹500, making the contract feel neutral to both parties; however, over the next three months, the prices could rise to ₹700 or fall to ₹400. When the delivery date arrives, and the market price has risen to ₹700, the buyer gains money because they are obliged to buy the bicycle at the pre-agreed price of ₹500, while the seller suffers a loss. Importantly, the gain or loss experienced by either party is not directly because of the cycle, but the contract based on the cycle that the parties agreed on, meaning that the contract derives its value from the underlying asset. The word “derivative” originates from the term “to derive,” which means that the value of the contract is determined indirectly because of something else.

A derivative is more technically defined as a financial contract whose value derives from the value of other, more basic, variables. Often, the variables that underlie derivatives are the prices of assets. But derivatives don’t have to be based on assets only; for example, weather derivatives are a financial contract

that companies use to hedge against potential losses that are derived from weather conditions.

Standardised contracts are traded in exchange-traded markets called derivatives exchanges. The first derivatives exchange was the Chicago Board of Trade (CBOT), established in 1848, which would make contracts based on the quantity and quality of grains. Nowadays, derivatives are traded on electronic markets, similar to stocks.



The derivatives market has become one of the largest financial markets all around the globe, with futures and options being very popular financial instruments, with the OTC market having grown to 630 trillion USD back in December 2012. Some derivatives are not traded in exchange-traded markets and are traded directly in over-the-counter markets (OTC Markets). The main participants in OTC markets are banks, financial institutions, and corporations. Usually, a financial intermediary called a central counterparty (CCP) is used to clear the trade bilaterally and reduce the risk either party in the trade has to bear.

Types of Derivatives

1. Forward Contracts:

A forward contract is an agreement to buy or sell an asset at a certain time in the future at a pre-defined price. It is differentiated from a spot contract, which is an agreement to buy or sell an asset immediately. Forward

contracts are traded in the OTC market, usually between two financial institutions or between a financial institution and a client. One party to a forward contract assumes a long position (agrees to buy the underlying asset at a specified future date, for a specified price), and the other assumes a short position (agrees to sell the asset on the same date for the same price). Forward contracts on foreign exchanges are particularly popular.

2. Futures Contracts:

A futures contract is similar to a forward contract, except for the fact that futures contracts are traded in exchange-traded markets, whereas forward contracts are traded in the OTC (Over the Counter). Exchanges standardise certain features on futures contracts, and hence, futures contracts are much less 'customizable' than forward contracts. A wide range of futures contracts based on financial assets and commodities are traded on the Chicago Board of Trade (CBOT). Gold, silver, and oil are some of the most popular commodity futures on exchanges worldwide. The payoff for a futures contract is:

$$\text{Long Position Payoff} = S - K$$

$$\text{Short Position Payoff} = K - S$$

where S is the spot price at maturity, and K is the delivery price

If we want to trade futures on an exchange, how would we price a futures contract? Since we have stated that, according to no-arbitrage, it's not possible to extract a risk-free profit, we should get the exact same cash flows from two different strategies for the same underlying asset.

If a trader enters a futures contract (whose underlying asset is currently priced at $S(t)$) at a time t , with maturity at T , priced at $F(t, T)$, the only cash flow is a payment of $F(t, T)$ at a time T .

Now, say the trader takes on another strategy and borrows $S(t)$ at a risk-free rate r , buys the underlying asset, and holds it till maturity. At maturity, they are both in the same position, owning the asset that is priced at $S(T)$. The trader owes $S(t)(1 + r)^{T-t}$ in this strategy.

According to no-arbitrage, both strategies should generate the same cash flow; hence, the price of a futures contract would be:

$$F(t, T) = S(t)(1 + r)^{T-t}$$

In the modern day, futures contracts are repriced continuously, and the P&L is constantly being transferred between the long holder and the short holder, rather than at the maturity period, which one might initially assume. Since the contract is being repriced continuously, the future's price would be modified as:

$$F(t, T) = S(t) \left(1 + \frac{r}{n}\right)^{n(T-t)}$$

where n is the compounding frequency

Since the futures contracts are being updated continuously, the compounding frequency becomes very large.

$$\lim_{n \rightarrow \infty} S(t) \left(1 + \frac{r}{n}\right)^{n(T-t)} = S(t)e^{r(T-t)}$$

This modifies the pricing formula to:

$$F(t, T) = S(t)e^{r(T-t)}$$

Practically, the pricing depends on many more factors than just the risk-free rate. Some underlying assets produce dividends, and yields might also be included.

Because the futures prices are being updated continuously, continuous P&L transfer between the long holder and the short holder occurs. But say one position holder does not have enough money to give to the counterparty (who, in turn, transfers it to the opposite position holder), then where will the counterparty get the money to transfer? The counterparty might have to bear the loss themselves. The long holder and short holder have to pay an initial collateral, called a margin, for this reason. This margin is a small portion of the contract price and is used to take losses and add profits continuously.

Say 2 market participants enter a BTC futures contract at \$50,000, and they both pay a margin of \$5,000. The counterparty has essentially let you enter a \$50,000 contract while paying only \$5,000. This is called leverage, and it amplifies your capital invested and, in turn, amplifies your risk (potential losses, which can be greater than initially invested capital). In this case, the leverage is 10x. All modern exchange-traded futures are leveraged.

A pivotal process in modern financial markets is 'Liquidation', which is the forced closure of a trader's leveraged position by an exchange. Exchanges require all traders to have a minimum amount remaining in their margin called a "Maintenance Margin". If the market goes against a leveraged trade, losses can build up fast. If the margin drops below the maintenance margin, the exchange forcibly closes your position to prevent further losses and to make back the borrowed money. This is called 'forced liquidation'.

3. Options Contracts:

Options contracts are traded on both OTC and exchange-traded markets. The largest stock options exchange in the world is the Chicago Board Options Exchange (CBOE). There are two types of options, viz., call (buy) options and put (sell) options. A call option gives the holder of the contract the right to buy the underlying asset by a certain date for a certain price; similarly, a put option gives the holder of the contract the right to sell the underlying asset by a certain date for a certain price. The price the contract goes for is called the strike price, and the date at which the contract expires is called the maturity.

American options give the holder the right to exercise the contract at any time up to maturity, whereas European options give the holder the right to exercise the contract only at the maturity date itself. The difference between the previous 2 derivative contracts and options contracts is that options contracts give the holder the right, but not the obligation, to exercise the contract. The holder need not exercise his right during the duration of the contract. Also, there is a cost to entering an options contract, whereas futures and forward contracts don't have any entrance cost.

Perpetual Futures & Funding Rates

Perpetual futures are derivative contracts that have the same features as traditional futures contracts, but they lack a maturity date. They allow traders to speculate at no cost on the price movements of an asset or a commodity.

The lack of a maturity date gives two main advantages to market participants. The first one being that there is no need to move from matured contracts to new contracts, and the second being that since there aren't various contracts with

different maturity periods now, liquidity increases as all traders can trade one single perpetual futures contract.

This allows for the market price to reflect all available information because all traders interact with the same type of contract, and the different views of traders (hedgers, arbitrageurs, speculators) are reflected.

Perpetual futures were first introduced by Robert Shiller in 1993. He introduced it as a solution to set a perpetual claim on the cash flows from an asset that was either illiquid or difficult to measure, for example, a house that generates rent as cash flows.

Essentially, Shiller envisioned holding one unit of a perpetual future contract valued at f_t at a time t would be equivalent to having a claim on a dividend stream (source of dividends) s_{t+1} of a financial instrument without ownership of the financial instrument, and without maturity.

Now, we long this perpetual futures contract, the cash flow from t to $t + 1$ denoted as s_{t+1} , which is transferred from the short holder to the long holder in the form of daily settlements, including the changes in the futures price. If the futures contract has become f_{t+1} at $t + 1$, the long holder gains $f_{t+1} - f_t$ and the short holder loses $f_{t+1} - f_t$, by virtue of the change in the futures contract price alone. So the effective cash flow the long holder receives from the short holder without the change in the futures price is $s_{t+1} - (f_{t+1} - f_t)$

Instead, the market participant could borrow f_t worth of cash at time t at a risk-free rate r_t , and use that cash to acquire a financial asset or a portfolio that produces a dividend stream d_{t+1} . He has to repay $f_t(1 + r_t)$ between t and

$t + 1$ as he has borrowed the cash, so effectively his cash flow from t to $t + 1$ is $d_{t+1} - r_t f_t$.

But, according to no arbitrage, these two strategies should lend the same cash flows to the market participants.

$$\Rightarrow s_{t+1} - (f_{t+1} - f_t) = d_{t+1} - r_t f_t$$

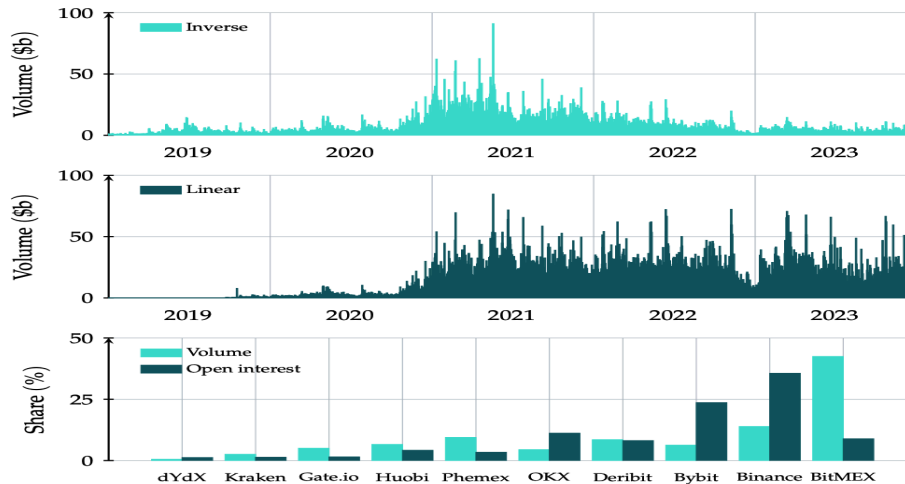
Thus, the daily settlement the short holder makes to the long holder is:

$$s_{t+1} = (f_{t+1} - f_t) + (d_{t+1} - r_t f_t)$$

Perpetual futures (also called perps) are particularly popular in the cryptocurrency markets, generating daily volumes of over \$100 billion, especially BTC/USD perpetual futures. They are soon expected to become popular among commodities, too.

Perpetual futures were introduced into the cryptocurrency market later on, but with a purpose different from what Shiller originally envisioned, since cryptocurrencies don't have any cash flows or dividends, and also, crypto spot prices can be measured on exchanges.

The major use of perpetual futures in the crypto markets was to offer an effective financial instrument to either hedge or speculate on the movement of the spot price. Perpetual futures were first introduced into the cryptocurrency markets in 2016 by BitMEX, and they have gained great popularity since then. According to CoinGecko, the median total daily trading volume of perpetual futures across all exchanges is ~102 billion in 2022, around 2-3x the total spot trading volume.



Now, perpetual futures can be classified based on their settlement asset. A settlement asset determines what you post as your margin, how you receive your settlement, and what types of risk you're exposed to.

1. Coin-Margined (Inverse) Perpetuals:

Inverse perpetual futures are contracts in which both the margin and P&L are denominated in the underlying crypto-asset, even though the contract is priced in USD. Usually, the contract is priced in the form of USD per BTC, for example.

The value of the contract follows an inverse relationship with the value of the underlying crypto asset. For example, if you buy a \$10,000 inverse perpetual contract when 1 BTC costs \$25,000, and you close the contract when 1 BTC costs \$30,000. Now with the \$10,000, you can buy back BTC and make 0.33 BTC, earning yourself a 0.07 BTC profit.

2. USDT (or) Fiat-Margined (Linear) Perpetuals:

Linear perpetual futures are contracts in which both the margin and P&L are denominated in stablecoins like USDT or USDC, while the contract price is quoted in the underlying crypto-asset.

The profit derived from the contract linearly varies with the value of the underlying crypto asset. For example, if a trader longs a 1BTC contract, and the value of 1 BTC rises by \$100, the trader makes \$100 profit. The majority of perpetual futures being traded in cryptocurrency markets are linear perpetuals, being the dominant instrument in the crypto-derivatives market.

Normally, in perpetual futures markets, the price of the contract should be close to the spot price of the underlying asset. However, due to supply and demand dynamics, perpetual futures are not guaranteed to converge to the spot price. This leads to premiums or discounts in the futures pricing. In traditional futures, the maturity date forces the contract price and the spot price to converge at maturity (due to no arbitrage), but due to the lack of an expiry date, this doesn't need to happen for perpetual futures.

Also, since crypto-assets like Bitcoin don't generate dividends ($d_{t+1} = 0$), the periodical settlement price is:

$$s_{t+1} = (f_{t+1} - f_t) - r_t f_t$$

The term $r_t f_t$ requires the risk-free rate of a short-period risk-free alternative crypto-asset, but what is the risk-free rate in the cryptocurrency market? Is there a crypto-asset that can generate consistent returns with zero scope of defaulting? In traditional financial markets, the risk-free rate is anchored to extremely safe sovereign bonds like U.S. Treasury bills. The cryptocurrency market lacks any type of sovereign guarantee or central bank that can print currency. So, how does no arbitrage hold for perpetual futures, and what's used to serve as $r_t f_t$?

To solve this issue, an important feature of crypto perpetual futures is the funding rate, which is a cash fee that gets exchanged between long position

holders and short position holders. It's meant to keep the futures price close to the underlying asset's spot price, to solve the no-arbitrage issue, and so that futures can stay an effective hedging tool.

The typical funding rate period is every 8 hours. If the gap between the futures price and the spot price is positive, then the long position holder has to pay the short position holder, which incentivises traders to short the future, which will then reduce the price gap and bring the price back to being close to the spot price. If the price gap is negative, then the opposite will happen.

Funding rates have 2 major components, one being the exchange-set interest rate, and the other component called the 'Clamp'.

$$\text{Funding Rate} = \text{Interest Rate} + \text{Clamp}$$

The interest rate component is decided by cryptocurrency exchanges, and the rate is determined depending on the exchange. One method is having a fixed baseline rate, which represents a theoretical risk-free rate. Another is to benchmark it to observable interest rates like the London Inter-bank Offered Rate (LIBOR) or Sovereign Overnight Financing Rate (SOFR), or sovereign bond yields.

The clamp is related to the 'Premium Index', which is the percentage by which the perpetual futures contract price varies from the spot price. The clamp effectively bounds the premium index component, usually between -0.05% and +0.05%. The reason behind this is that during times of high volatility or low liquidity, this cap protects traders from excessively high funding rates. Without the clamp, funding rates wouldn't be capped, and this could lead to mass liquidation.

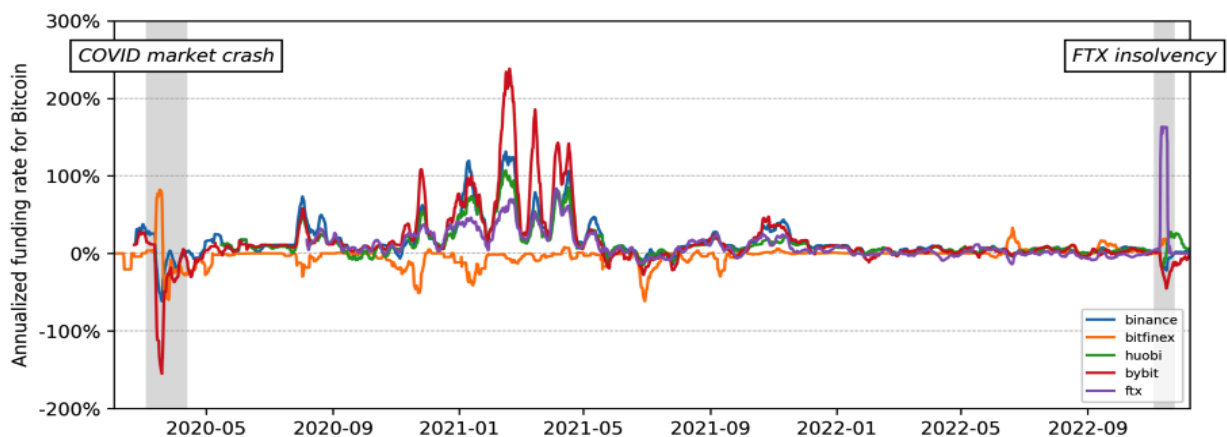
Once the funding rate is calculated as the sum of the interest rate and the clamped premium component, it affects trader incentives through periodic

payments between long and short positions. When the perpetual contract trades above the spot price, the premium index becomes positive and, after clamping, produces a positive funding rate, thus longs to pay shorts. This increases the cost of holding long positions and makes traders reduce longs or open shorts. The resulting bearish pressure pushes the perpetual price downward toward the spot index, reducing the premium index used in subsequent funding calculations. At the simplest level, entering a perpetual futures contract at time t will give periodic payments at $t + 1$ equal to:

$$s_{t+1} = (f_{t+1} - f_t) - k(f_t - x_t)$$

where $k > 0$ and x_t is the spot price of the underlying asset at time t .

As the premium narrows, the funding payment component decreases, lowering the funding rate and weakening the incentive to short. When the perpetual trades below spot, the premium index is negative, and funding payment becomes negative, causing shorts to pay longs. This encourages long positions, increasing bullish pressure and lifting the perpetual price toward spot. As the price deviation shrinks, the premium index declines in magnitude, reducing the clamp contribution. Thus, funding rates create a feedback loop system that maintains the perpetual spot and spot price.



The above picture shows the 7-day moving average annualised funding rate for BTC between 2020 and 2023. Two key events here are the COVID stock market crash in early 2020 and the FTX bankruptcy in November 2022. The FTX

bankruptcy caused a brief period of significantly negative funding rates on other solvent exchanges, resulting in perpetual futures being priced lower than the spot prices.

Squeezes in the Cryptocurrency Market

In cryptocurrency markets, 'squeezes' refer to market events where a rapid and sharp change in the price of crypto-assets occurs. This causes traders who bet against the change to square off their position to cover their position, which further drives the change. There are two types of squeezes, viz. Short squeeze and Long squeeze.

1. Short Squeeze:

A short squeeze is a rapid, sharp increase in the price of a crypto-asset. It occurs when a crypto-asset's price rises unexpectedly, forcing traders with short positions to square off their positions by purchasing the crypto-asset, which in turn causes the price of the crypto-asset to become even higher.



A short squeeze starts with an accumulation of short positions, with traders hoping for the price of the crypto-asset to fall, and buy it back at a lower price. The trigger for the short squeeze happens when an event

occurs that causes the price of the crypto-asset to rise rapidly. This could be due to a change in sentiment about the market or unexpected positive news about the asset. As the price increases, short sellers begin to incur losses.

As we know, perpetual futures are heavily leveraged (up to 100x or more sometimes), so this means even minor movements in the price could lead to liquidation. So, to cover their position or due to forced liquidation, traders are forced to sell their assets to square off their position.

This leads to a substantial demand in the market for this asset, which further accelerates the price up. This higher price triggers more forced liquidations, further increasing the price and creating a positive feedback loop.

2. Long Squeeze:

A long squeeze is a rapid drop in the price of a crypto-asset. It occurs when the price of the asset drops suddenly, forcing traders to square off their long positions by selling their asset, in turn leading to an even lower price. Long squeezes are much rarer than short squeezes because, in a short squeeze, holders of short positions of spot assets are obligated to settle the shorted assets, whereas no such obligation exists for long position holders.

A long squeeze starts with an accumulation of long positions. Traders hope for the price of the crypto-asset to rise, and sell it to extract a profit. The trigger for a long squeeze could be sudden negative news about the asset, a change in a regulatory policy, or just negative market sentiment.

As the price drops, long holders, especially those who are in heavily leveraged long perpetual futures positions, start to reach their margin

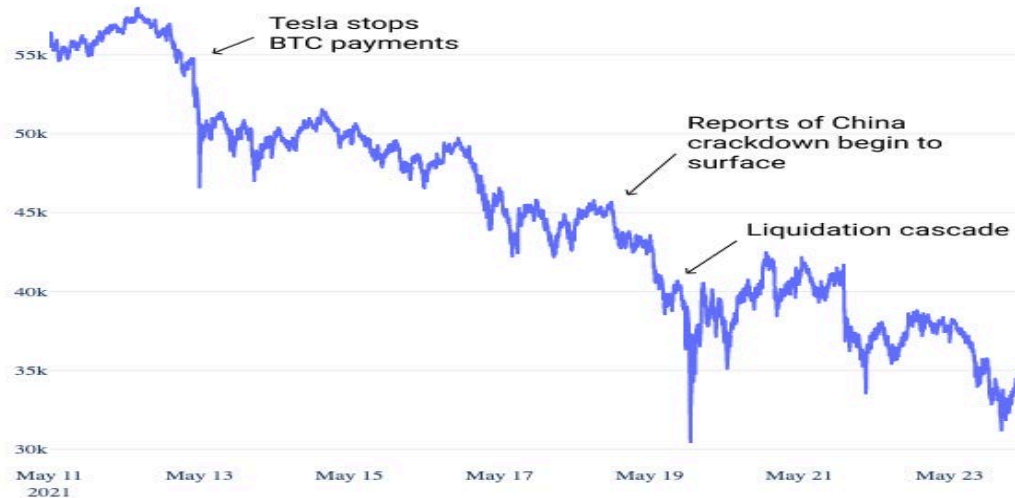
threshold. They have to start selling their asset, or else their asset will be forcibly liquidated.

This leads to a large supply of the asset in the market, driving the price down even faster. This triggers more liquidation and selling, further decreasing the price, and creating a positive feedback loop.

The 2021 Bitcoin Crash: A Case Study

BTC Price, 5/11/21-5/23/21

Source: Coin Metrics Reference Rates



The 2021 Bitcoin long squeeze is one of the best case studies to understand how volatile the derivatives market is and how leverage & liquidation work. The sharp decline that occurred in May 2021 wasn't just due to some bad news regarding Bitcoin, but was also because of the structure of the cryptocurrency derivatives market. The event demonstrates how leverage, funding payment mechanisms, and automated liquidation systems can interact to generate self-reinforcing price dynamics.

Months before the crash, BTC was priced at its peak and was giving high returns. It rose from USD 20,000 in late 2020 to USD 65,000 in April 2021. Funding rates

during this period were highly positive, which indicated that long positions were dominating the market and that traders were willing to pay a premium to maintain their leveraged bullish position. The triggers that led to this cascade were announcements related to restrictions on cryptocurrency mining and trading in China, alongside heightened scrutiny of Bitcoin's environmental impact. This was just the trigger for the fall, but not the main reason for it. Once the prices began to decline, leveraged long positions started to reach their margin threshold, and since at that moment the leverage was at 100-200x, a small decline in the price caused huge liquidation of perpetual futures. Due to liquidation, the prices kept going lower, causing more and more sell orders. Thus leading to a cycle of crashes.

During May 2021, a billion dollars' worth of long positions were liquidated within a very short timeframe, producing sharp intraday price drops. A critical feature of this cascade was the key role that the derivatives market played in the price discovery of Bitcoin. Often in the cryptocurrency market, perpetual futures exhibit greater liquidity and faster reaction times than spot markets. This caused funding rates to not be able to prevent the decline of the crash, because it happened very fast, quicker than the funding rate period. The funding payments did manage to turn negative later, but it was too late. The mismatch between risk accumulation and risk correction, which should've been solved by the funding rate mechanism, amplified volatility rather than dampening it.

Notably, the market later recovered and reached a new all-time high in November 2021, which says that the crash was not a test of Bitcoin's long-term value but rather a deleveraging event driven by the crypto-derivatives market structure and volatility of derivatives.

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